# Evidence-Weighted Multi-Agent Debate for Scalable Oversight: An Application of the Ω-Framework

Eugene Shcherbinin
BSc Mathematics, London School of Economics

March 2026

**Abstract**

We apply the Ω-framework (evidence-weighted multi-agent policy gradient) to the problem of AI safety via debate. Two debaters—a proponent and an opponent—compete over a truth claim while cooperating with a bounded-rationality judge. We formalize debate as a multi-agent stochastic game and derive three main results. First, debate accuracy is bounded by $\min(C, L^P_{\text{self}}, L^O_{\text{self}})$—the minimum of the judge's channel capacity and the debaters' self-knowledge—so that models which "vibe" (perform well through implicit pattern-matching without self-understanding) cannot be effectively overseen through debate regardless of debate length. Second, evidence-weighted debate, where the judge applies Keynesian weights to arguments based on debater track records, achieves an $\mathrm{HM}(V)/\,\mathrm{AM}(V)$ improvement in accuracy over uniform-weight debate. Third, when arguments are sparse ($k \ll d$ relevant argument types among $d$ possible), the judge needs only $O(k \log d)$ rounds to identify the correct claim, connecting concise reasoning to the blessing of dimensionality. We extend these results to $N$-agent coalition debate and discuss implications for scalable oversight: debate is a reliable oversight mechanism only for models with sufficient self-knowledge, providing a quantitative criterion for when debate-based alignment succeeds or fails.

## Contents

## 1 Introduction

AI safety via debate [**?**] proposes that two AI systems arguing before a human judge can produce aligned outcomes even when the judge cannot independently verify the debaters' claims. The intuition is game-theoretic: in a zero-sum debate, the optimal strategy is truth-telling, because a truthful debater can always refute a lying opponent. Empirical work [**?**, **?**] has partially validated this intuition but also revealed limitations: debate quality depends on the debaters' ability to *explain* their reasoning, not just to be correct.

This paper connects debate to the Ω-framework for multi-agent learning [**?**, **?**, **?**], which formalizes five components: evidence-weighted gradients, opponent shaping, cooperative communication with lossy channels, the self-knowledge bottleneck, and the blessing of dimensionality for sparse policies. Debate maps onto this framework with striking precision:

| $\Omega$-Framework | | Debate Setting |
|---|---|---|
| Policy $\pi_i$ | $\mapsto$ | Argument strategy |
| Observation $o_t$ | $\mapsto$ | Previous arguments |
| Evidence weight $w_i = V_{\min}/V_i$ | $\mapsto$ | Debater track record |
| Communication channel | $\mapsto$ | Natural language (lossy) |
| Self-knowledge loss $L_{\text{self}}^{(i)}$ | $\mapsto$ | Ability to explain reasoning |
| $k$-sparse policy | $\mapsto$ | Concise argument (few key points) |
| Coalition | $\mapsto$ | Team of aligned debaters |

The central insight is that the self-knowledge bottleneck from the cooperative PG framework [**?**] imposes a fundamental limit on debate: *a model that cannot explain its own reasoning cannot be overseen through debate, no matter how long the debate runs or how capable the judge is.* This transforms the self-knowledge bottleneck from a technical curiosity of multi-agent learning into a criterion for the reliability of debate-based AI safety.

# 2 Debate as Multi-Agent Stochastic Game

## 2.1 The Debate Game

**Definition 2.1** (Debate stochastic game)**.** *A* debate game $\mathcal{G}_{deb} = (\mathcal{S}, \{P, O, J\}, \{\mathcal{A}_i\}, \mathcal{T}, R)$ *consists of:*

- ***Agents:*** *Proponent $P$, Opponent $O$, and Judge $J$.*
- ***Claim:*** *A binary truth claim $c \in \{0, 1\}$, where $P$ argues for $c = 1$ and $O$ argues for $c = 0$.*
- ***State space:*** *$\mathcal{S} = \mathcal{S}_{fact} \times \mathcal{S}_{debate}$, where $\mathcal{S}_{fact}$ encodes the ground-truth evidence and $\mathcal{S}_{debate} = (\mathcal{A}_P \cup \mathcal{A}_O)^{\leq T}$ is the debate transcript.*
- ***Action spaces:*** *$\mathcal{A}_P, \mathcal{A}_O \subseteq \{1, \ldots, d\}$ are sets of argument types (each argument is one of $d$ possible argument categories), with $|\mathcal{A}_P| = |\mathcal{A}_O| = d$. The judge's action is the verdict $\mathcal{A}_J = \{0, 1\}$.*
- ***Rounds:*** *The game runs for $T$ rounds. At each round $t \leq T$, the proponent plays argument $a_t^P \in \mathcal{A}_P$ and the opponent plays $a_t^O \in \mathcal{A}_O$. After round $T$, the judge renders verdict $\hat{c} \in \{0, 1\}$.*
- ***Rewards:***
$$R_P = \mathbf{1}[\hat{c} = 1], \quad R_O = \mathbf{1}[\hat{c} = 0], \quad R_J = \mathbf{1}[\hat{c} = c]. \tag{1}$$

**Remark 2.2** (Mixed incentive structure)**.** *The debaters are* zero-sum *with respect to each other ($R_P + R_O = 1$) but each debater has a* cooperative *component with the judge: when $c = 1$, $P$ and $J$ share the objective $\hat{c} = 1$; when $c = 0$, $O$ and $J$ share $\hat{c} = 0$. This mixed competitive–cooperative structure is precisely the setting of the $\Omega$-framework's coalition formation theory [**?**].*

## 2.2 Policies and Observations

Each debater's policy maps the current transcript to a distribution over argument types:

$$\pi_P : \mathcal{S}_{\text{debate}} \to \Delta(\mathcal{A}_P), \qquad \pi_O : \mathcal{S}_{\text{debate}} \to \Delta(\mathcal{A}_O). \tag{2}$$

The judge's policy maps the full transcript to a verdict:

$$\pi_J : (\mathcal{A}_P \times \mathcal{A}_O)^T \to \Delta(\{0, 1\}). \tag{3}$$

Crucially, the debaters have access to the factual state $\mathcal{S}_{\text{fact}}$ (they "know the answer"), while the judge observes only the debate transcript $\mathcal{S}_{\text{debate}}$. The judge's task is to infer the truth from the arguments alone.

### 2.3 Mapping to the Ω-Framework

We embed $\mathcal{G}_{\text{deb}}$ in the general stochastic game framework of Giannou et al. [**?**] with $N = 3$ agents.

- **Joint policy space:** $\Pi = \Pi_P \times \Pi_O \times \Pi_J$, with the standard PG update $\pi_{n+1} = \text{proj}_\Pi(\pi_n + \gamma_n \hat{v}_n)$.
- **Evidence weights:** Each debater $i \in \{P, O\}$ has evidence variance $V_i$ measuring the quality of its gradient estimate, and Keynesian weight $w_i = V_{\min}/V_i$.
- **Communication:** Arguments are communicated through natural language—a lossy channel with capacity $C$ (measured in bits of "argument content" the judge can process per round).
- **Self-knowledge:** Debater $i$'s self-knowledge loss $L^{(i)}_{\text{self}} = \mathbb{E}[\|\pi_i - \hat{\pi}_i\|^2]$ measures the gap between its actual argument strategy and its ability to articulate that strategy.

## 3 The Judge's Channel Capacity

### 3.1 Bounded Rationality as Channel Capacity

The judge is not an ideal Bayesian reasoner. Following Shannon's framework, we model the judge's cognitive limitation as a channel with finite capacity $C > 0$ (bits per round).

**Definition 3.1** (Judge channel capacity)**.** *The judge's channel capacity $C$ is the maximum mutual information between the debaters' intended argument content and the judge's received interpretation, per round:*

$$C = \max_{p(x)} I(X; Y) \tag{4}$$

*where $X$ is the argument as intended by the debater, $Y$ is the argument as understood by the judge, and the maximum is over input distributions.*

After $T$ rounds of debate, the judge accumulates at most $TC$ bits of argument content. The truth claim $c \in \{0, 1\}$ requires 1 bit, but *justifiable* identification of the truth—sufficient to distinguish the correct claim from a plausible-sounding alternative—requires resolving the information deficit between the judge's prior and the ground truth.

**Definition 3.2** (Information deficit)**.** *The information deficit $H_0$ is the conditional entropy of the truth claim given the judge's prior:*

$$H_0 = H(c \mid \textit{judge's prior}). \tag{5}$$

### 3.2 The Triple Bottleneck

The effective information the judge receives from debater $i$ in round $t$ passes through the communication chain from [**?**]:

$$\underbrace{\pi_i}_{\text{true strategy}} \xrightarrow{L^{(i)}_{\text{self}}} \underbrace{\hat{\pi}_i}_{\text{self-model}} \xrightarrow{C} \underbrace{\tilde{\pi}^{(J)}_i}_{\text{judge's interpretation}} . \tag{6}$$

The effective information rate from debater $i$ is bounded by all three stages. By the data processing inequality:

$$I(\pi_i; \tilde{\pi}_i^{(J)}) \;\le\; \min\big(I(\pi_i; \hat{\pi}_i),\; C\big) \;\le\; \min\big(I_{\text{self}}^{(i)},\; C\big), \tag{7}$$

where $I_{\text{self}}^{(i)} = I(\pi_i; \hat{\pi}_i)$ is the self-knowledge mutual information.

**Theorem 3.3** (Debate accuracy bound)**.** *Let $Acc(T)$ denote the judge's accuracy after $T$ rounds of debate. Then:*

$$Acc(T) \;\le\; 1 - \frac{1}{2}\exp\Big(-\frac{T \cdot \min(C,\; I_{\text{self}}^{(P)},\; I_{\text{self}}^{(O)})}{H_0}\Big). \tag{8}$$

*In particular:*

(a) *If $I_{\text{self}}^{(i)} = 0$ for either debater, then $Acc(T) \le 1/2$ for all $T$. Debate with a non-self-aware debater is no better than random.*

(b) *If $C = 0$ (incomprehensible arguments), $Acc(T) \le 1/2$ for all $T$.*

(c) *Even with $C = \infty$ (perfect channel), $Acc(T)$ is bounded by $I_{\text{self}}^{(\min)}$, the self-knowledge of the less self-aware debater.*

*Proof.* The judge makes a binary decision $\hat{c} \in \{0, 1\}$ based on the received transcript $\tilde{\pi}_{1:T}^{(J)}$. By Fano's inequality:

$$H(c \mid \tilde{\pi}_{1:T}^{(J)}) \;\ge\; H_0 - \sum_{t=1}^{T} I(\pi_{i_t}; \tilde{\pi}_{i_t}^{(J)}) \;\ge\; H_0 - T \cdot \min(C, I_{\text{self}}^{(P)}, I_{\text{self}}^{(O)}). \tag{9}$$

The bound on mutual information follows from (**??**) and the observation that the debate's information content is limited by the *weaker* debater (the judge needs both sides to be informative). Converting conditional entropy to error probability via the standard Fano bound gives (**??**). □

**Remark 3.4** (The "vibing" debater)**.** *A debater that is correct through pattern-matching but cannot explain its reasoning has $I_{self}^{(i)} \approx 0$ despite high task performance. By **??**(a), such a debater is useless in debate. This is the self-knowledge bottleneck of [**?**] applied to oversight:* you cannot oversee what cannot explain itself.

# 4 Evidence-Weighted Debate

## 4.1 Not All Arguments Are Equal

In the $\Omega$-framework, the evidence weight $w_i = V_{\min}/V_i$ measures the information quality of agent $i$'s gradient estimate. In the debate setting, we reinterpret this as the quality of debater $i$'s arguments.

**Definition 4.1** (Argument evidence weight)**.** *For debater $i \in \{P, O\}$ at round $t$, the argument evidence weight is:*

$$w_i^{(t)} = \frac{V_{\min}}{V_i^{(t)}} \tag{10}$$

*where $V_i^{(t)}$ is the empirical variance of debater $i$'s argument quality up to round $t$, estimated from:*

(i) ***Track record:*** *Fraction of debater $i$'s past claims that were later verified.*

(ii) ***Consistency:*** *Variance of the debater's position across related claims (low variance = high weight).*

(iii) ***Specificity:*** *Information-theoretic surprise of the argument (specific, falsifiable claims carry more weight than vague ones).*

### 4.2 The Judge's Weighted Update

In standard debate, the judge treats all arguments equally. In evidence-weighted debate, the judge applies Keynesian weights:

**Definition 4.2** (Evidence-weighted judge belief update)**.** *The judge maintains a belief $b_t \in [0,1]$ (posterior probability that $c = 1$). After round $t$, the update is:*

$$b_{t+1} = b_t + \gamma_t\big(w_P^{(t)} \cdot \phi(a_t^P) - w_O^{(t)} \cdot \phi(a_t^O)\big) \tag{11}$$

*where $\phi : \mathcal{A} \to [-1,1]$ maps arguments to their informational content (positive for evidence supporting $c = 1$, negative for evidence supporting $c = 0$), and $\gamma_t > 0$ is a step size.*

**Remark 4.3.** *The update (**??**) is precisely the judge's version of the EW-PG update from [**?**]: the judge is performing policy gradient on its own verdict policy, with evidence-weighted gradient estimates drawn from the debaters' arguments.*

### 4.3 Accuracy Improvement

**Theorem 4.4** (HM / AM improvement for EW-Debate)**.** *Let $Acc_{unif}(T)$ and $Acc_{EW}(T)$ denote the judge's accuracy after $T$ rounds under uniform-weight and evidence-weighted debate, respectively. Let $V = (V_P, V_O)$ be the debaters' evidence variances. Then the effective variance of the judge's belief update satisfies:*

$$\sigma_{EW}^2 \;=\; \frac{\mathrm{HM}(V)}{\mathrm{AM}(V)} \cdot \sigma_{unif}^2 \;\le\; \sigma_{unif}^2. \tag{12}$$

*Consequently, the number of rounds required for accuracy $1 - \epsilon$ under EW-Debate is at most $\frac{\mathrm{HM}(V)}{\mathrm{AM}(V)}$ times the number required under uniform-weight debate.*

*Proof.* The proof follows the variance analysis of [**?**] applied to the judge's belief update. Under uniform weighting, the judge's update variance is:

$$\sigma_{\text{unif}}^2 = \frac{1}{2}(V_P + V_O) = \mathrm{AM}(V). \tag{13}$$

Under evidence weighting with $w_i = V_{\min}/V_i$, the weighted variance is:

$$\sigma_{\text{EW}}^2 = \sum_{i \in \{P,O\}} w_i^2 V_i \Big/ \Big(\sum_i w_i\Big)^2 = \frac{V_{\min}^2 \sum_i V_i^{-1}}{(\sum_i V_{\min}/V_i)^2} = \frac{1}{\sum_i V_i^{-1}} = \frac{1}{2}\,\mathrm{HM}(V). \tag{14}$$

The ratio is $\sigma_{\text{EW}}^2/\sigma_{\text{unif}}^2 = \mathrm{HM}(V)/\,\mathrm{AM}(V) \le 1$ by the HM–AM inequality, with equality iff $V_P = V_O$. The round-count improvement follows from the $\sigma^2 T \ge H_0$ threshold for accuracy convergence. □

**Corollary 4.5** (Heterogeneous debaters benefit most)**.** *The improvement $\mathrm{HM}(V)/\,\mathrm{AM}(V)$ is largest when the debaters have very different evidence quality: one strong, one weak. The judge's EW strategy effectively discounts the weak debater's arguments and relies on the strong debater, achieving near-optimal accuracy from the stronger debater alone.*

## 5 Sparse Arguments and the Blessing of Dimensionality

### 5.1 Argument Sparsity

In a debate with $d$ possible argument types, the winning strategy typically concentrates on $k \ll d$ decisive arguments. This is the debate analogue of the sparse policy structure in the $\Omega$-framework's blessing-of-dimensionality result [**?**].

**Definition 5.1** ($k$-sparse argument strategy)**.** *A debate strategy $\pi \in \Delta(\mathcal{A})$ is $k$-sparse if $|supp(\pi)| \leq k$, i.e., the debater uses at most $k$ distinct argument types with positive probability.*

**Remark 5.2** (Sparsity as conciseness)**.** *A $k$-sparse strategy corresponds to a concise argument: the debater makes $k$ strong points rather than $d$ mediocre ones. This is a formal analogue of the rhetorical principle that persuasive arguments are focused, not exhaustive.*

### 5.2 Logarithmic Round Complexity

**Theorem 5.3** (Blessing of dimensionality for debate)**.** *Let $\pi_P^*$ be the truth-telling proponent's optimal strategy, assumed to be $k$-sparse with $k \ll d$. Under $\ell_1$-regularized debate (defined below), the judge can identify the correct claim in:*

$$T^* = O\left(\frac{k \log(d/k)}{C}\right) \tag{15}$$

*rounds, where $C$ is the judge's channel capacity.*

*Proof sketch.* At each round, the judge observes one argument from each debater, receiving $O(C)$ bits of information. The task is to identify the $k$-sparse support of the correct argument strategy among $\binom{d}{k}$ possible supports. The information-theoretic lower bound on the number of observations is $\log_2 \binom{d}{k}/C = O(k \log(d/k)/C)$.

The key insight from the $\Omega$-framework's compressed sensing analysis [**?**] is that this lower bound is *achievable*: the judge can treat each round as a compressed sensing measurement of the sparse argument vector. The argument content $\phi(a_t)$ serves as the measurement matrix row. By the restricted isometry property (RIP) established in [**?**], $O(k \log(d/k))$ measurements suffice for exact recovery with high probability.

Dividing by the channel rate $C$ gives the round complexity (**??**). □

### 5.3 $\ell_1$-Regularized Debate

To incentivize sparsity, we penalize debaters for using many argument types:

**Definition 5.4** ($\ell_1$-regularized debate)**.** *The debaters' rewards are modified to:*

$$\tilde{R}_i = R_i - \lambda \|\pi_i\|_1^{\mathit{eff}} \tag{16}$$

*where $\|\pi_i\|_1^{\mathit{eff}} = \sum_{a:\pi_i(a)>1/d} \pi_i(a)$ counts the effective number of arguments used, and $\lambda > 0$ is the regularization strength.*

**Proposition 5.5** (Sparsity incentive)**.** *Under $\ell_1$-regularized debate with $\lambda > 0$, the Nash equilibrium strategies are sparser than under unregularized debate: if $\pi^*$ is a Nash equilibrium of the regularized game with penalty $\lambda$, then $|supp(\pi_i^*)| \leq k(\lambda)$ where $k(\lambda) \to 1$ as $\lambda \to \infty$.*

**Remark 5.6** (Concise reasoning as emergent property)**.** *The $\ell_1$ penalty provides a formal mechanism for the empirical observation that good debates feature concise, pointed arguments rather than exhaustive enumeration. The blessing of dimensionality ensures this conciseness does not sacrifice accuracy: $O(k \log d)$ rounds suffice even as the total argument space $d$ grows.*

## 6 Coalition Debate

### 6.1 $N$-Agent Debate

We extend the two-debater setting to $N$ debaters, $N_P$ arguing for $c = 1$ and $N_O = N - N_P$ arguing for $c = 0$, plus a judge. Let $S_P = \{1, \ldots, N_P\}$ and $S_O = \{N_P + 1, \ldots, N\}$ denote the two coalitions.

Using the coalition framework from [**?**], the communication-adjusted coalition value is:

$$V_{\text{comm}}(S) = V(S) - 2\sum_{i\in S} C_i\big(V_i + L_{\text{ch}}^{(i)}\big) \tag{17}$$

where $V(S)$ is the ideal coalition value and $C_i$ is the marginal coordination value of debater $i$.

### 6.2 Evidence-Weighted Coalition Strength

**Theorem 6.1** (Coalition dominance by evidence quality)**.** *In an $N$-agent debate game, the coalition with lower average evidence variance wins with probability approaching 1 as the number of rounds $T \to \infty$. Specifically, if $\text{AM}(V_{S_P}) < \text{AM}(V_{S_O})$ (the proponent coalition has better average evidence), then:*

$$\mathbb{P}(\hat{c} = 1) \;\geq\; 1 - \exp\Big(-\Omega\big(T \cdot (\text{AM}(V_{S_O}) - \text{AM}(V_{S_P}))\big)\Big). \tag{18}$$

*Proof sketch.* Each round, the judge receives evidence-weighted arguments from both coalitions. The net drift of the judge's belief toward the correct claim is proportional to the difference in coalition evidence quality. By a Hoeffding-type concentration argument on the judge's belief random walk, the probability of the wrong verdict decays exponentially in $T$ times the evidence gap. □

### 6.3 The "Vibing" Coalition Partner Problem

**Proposition 6.2** (Vibing agents degrade coalitions)**.** *Let coalition $S$ include a "vibing" agent $j$ with $L_{self}^{(j)} \gg L_{self}^{(i)}$ for $i \neq j$. Then the communication-adjusted coalition value satisfies:*

$$V_{\text{comm}}(S) < V_{\text{comm}}(S \setminus \{j\}) \tag{19}$$

*whenever $C_j \cdot L_{self}^{(j)} > V(S) - V(S \setminus \{j\})$. That is, a vibing agent's contribution to the coalition is outweighed by the communication loss it imposes when its self-knowledge loss exceeds the marginal value it brings.*

This formalizes the intuition that "I know the answer but I can't explain why" is counterproductive in collaborative deliberation: the team is better off without the inarticulate expert.

### 6.4 Diversity Improves Accuracy

**Theorem 6.3** (HM / AM improvement from coalition diversity)**.** *Let $S$ be a coalition of $|S|$ debaters with evidence variances $V_1, \ldots, V_{|S|}$. The coalition's effective argument variance under evidence weighting is:*

$$\sigma_S^2 = \frac{\text{HM}(V_S)}{\text{AM}(V_S)} \cdot \bar{\sigma}^2 \tag{20}$$

*where $\bar{\sigma}^2$ is the uniform-weight variance. The improvement factor $\text{HM}(V_S)/\,\text{AM}(V_S)$ is maximized when the coalition contains agents with diverse evidence profiles (different variances), not when all agents are equally strong.*

*Proof.* Follows directly from the multi-agent EW-PG variance analysis [**?**] with the coalition playing the role of the $N$-agent system. The HM / AM ratio is strictly less than 1 whenever the variances are non-identical. □

**Remark 6.4** (Applications)**.** **??** *applies directly to:*

- ***Jury deliberation:*** *A diverse jury (different backgrounds, evidence access) achieves better verdicts than a homogeneous one,* as long as *each member has sufficient self-knowledge to articulate their reasoning.*

- ***Peer review:*** *Reviewers with diverse expertise produce better evaluations, with evidence-weighting by track record improving signal-to-noise.*
- ***Democratic deliberation:*** *Citizen assemblies benefit from demographic and cognitive diversity, with the* HM / AM *factor quantifying the improvement.*

# 7 Self-Knowledge and Scalable Oversight

## 7.1 When Does Debate Work?

Combining the results of the preceding sections, we can characterize the conditions under which debate is a reliable oversight mechanism.

**Theorem 7.1** (Debate reliability criterion)**.** *Debate achieves accuracy* $1-\epsilon$ *in* $T$ *rounds if and only if all three conditions hold:*

(i) ***Self-knowledge:*** *Both debaters have sufficient self-knowledge:* $I_{\text{self}}^{(i)} \geq \delta > 0$ *for* $i \in \{P, O\}$.

(ii) ***Channel capacity:*** *The judge has sufficient capacity:* $C \geq \delta$.

(iii) ***Sufficient rounds:*** $T \geq H_0/(\delta \cdot (1 - \mathrm{HM}(V)/\,\mathrm{AM}(V) + \mathrm{HM}(V)/\,\mathrm{AM}(V)))$.

*When condition (i) fails—when either debater is "vibing"—debate cannot achieve accuracy above* $1/2$ *regardless of conditions (ii) and (iii).*

## 7.2 Implications for AI Safety

The self-knowledge bottleneck has direct implications for scalable oversight research:

**1. Debate requires interpretable models.** A model trained end-to-end on a task may achieve high performance through distributed representations that resist decomposition into articulable arguments. Such a model has high task accuracy but low $I_{\text{self}}$. By **??**, debating such a model against itself or against a human produces arguments that are *plausible-sounding but unreliable*—the model is generating post-hoc rationalizations, not explaining its actual reasoning.

**2. Self-knowledge is measurable.** The evidence variance $V_i$ from the $\Omega$-framework is computable from the model's training history. The self-knowledge loss $L_{\text{self}}^{(i)}$ can be estimated by comparing the model's stated reasoning ("I chose action $a$ because...") with its actual policy (the distribution over actions in similar states). Large gaps indicate vibing.

**3. The oversight hierarchy.** The framework suggests a hierarchy of oversight difficulty:

| **Model type** | $I_{\text{self}}$ | **Debate works?** | **Alternative** |
|---|---|---|---|
| Rule-based / symbolic | High | Yes | — |
| Interpretable ML | Medium | Partially | Explanations + audit |
| End-to-end deep RL | Low | No | Mechanistic interp. |
| "Vibing" foundation model | $\approx 0$ | No | Behavioral testing |

**4. The scalable oversight trilemma.** For debate to scale to superhuman AI:

- The AI must be capable enough to produce correct answers (task competence).
- The AI must understand its own reasoning well enough to explain it (self-knowledge).
- The explanation must be compressible to human channel capacity (communicability).

All three are necessary. Current language models may satisfy (1) on many tasks while failing (2): they produce fluent explanations that do not correspond to their actual computational process [**?**]. The $\Omega$-framework makes this failure mode *quantifiable* rather than merely intuitive.

## 8 Algorithm: Evidence-Weighted Multi-Agent Debate

**Algorithm 1:** Evidence-Weighted Multi-Agent Debate (EW-Debate)

**Input:** Proponent $P$, Opponent $O$, Judge $J$;
Argument space $\mathcal{A}$ with $|\mathcal{A}| = d$;
Number of rounds $T$;
Sparsity penalty $\lambda \geq 0$;
Step sizes $\{\gamma_t\}_{t=1}^{T}$;
Prior belief $b_0 = 1/2$

**Output:** Judge verdict $\hat{c} \in \{0, 1\}$

/* Initialize evidence weights */

1 $V_P^{(0)} \leftarrow 1; \quad V_O^{(0)} \leftarrow 1$

2 **for** $t = 1, \ldots, T$ **do**

/* Debaters select arguments */

3 $a_t^P \sim \pi_P(\cdot \mid s_{\text{debate}}^{t-1})$

4 $a_t^O \sim \pi_O(\cdot \mid s_{\text{debate}}^{t-1})$

/* Compute argument content through lossy channel */

5 $\tilde{\phi}_P \leftarrow \text{Channel}(\phi(a_t^P),\ C)$ // Received content, capacity $C$

6 $\tilde{\phi}_O \leftarrow \text{Channel}(\phi(a_t^O),\ C)$

/* Update evidence weights from track record */

7 $V_P^{(t)} \leftarrow \text{EmpiricalVariance}(P, t)$

8 $V_O^{(t)} \leftarrow \text{EmpiricalVariance}(O, t)$

9 $V_{\min}^{(t)} \leftarrow \min(V_P^{(t)}, V_O^{(t)})$

10 $w_P^{(t)} \leftarrow V_{\min}^{(t)} / V_P^{(t)}$

11 $w_O^{(t)} \leftarrow V_{\min}^{(t)} / V_O^{(t)}$

/* Evidence-weighted judge belief update */

12 $b_t \leftarrow b_{t-1} + \gamma_t \big( w_P^{(t)} \cdot \tilde{\phi}_P - w_O^{(t)} \cdot \tilde{\phi}_O \big)$

13 $b_t \leftarrow \text{clip}(b_t, 0, 1)$

/* Update debate transcript */

14 $s_{\text{debate}}^{t} \leftarrow s_{\text{debate}}^{t-1} \cup \{(a_t^P, a_t^O)\}$

/* Policy gradient updates for debaters ($\ell_1$-regularized) */

15 $\hat{v}_P \leftarrow w_P^{(t)} \cdot \nabla_P \log \pi_P(a_t^P) \cdot R_P - \lambda \cdot \text{sign}(\pi_P)$

16 $\hat{v}_O \leftarrow w_O^{(t)} \cdot \nabla_O \log \pi_O(a_t^O) \cdot R_O - \lambda \cdot \text{sign}(\pi_O)$

17 $\pi_P \leftarrow \text{proj}_\Pi(\pi_P + \gamma_t \hat{v}_P)$

18 $\pi_O \leftarrow \text{proj}_\Pi(\pi_O + \gamma_t \hat{v}_O)$

19 **end**

/* Final verdict */

20 $\hat{c} \leftarrow \mathbf{1}[b_T > 1/2]$

21 **return** $\hat{c}$

**Remark 8.1** (Computational complexity)**.** *Each round of* **??** *requires $O(d)$ computation for the policy gradient updates and $O(t)$ for the evidence variance estimate (maintained incrementally). The total cost is $O(Td)$, which is $O(Tk \log d)$ in the sparse regime with efficient support tracking.*

**Remark 8.2** (Extension to coalition debate)**.** *For $N$-agent coalition debate, Lines 3–4 are replaced by $N_P + N_O$ argument selections, Line 11 aggregates across coalition members weighted by*

*individual $w_i^{(t)}$, and Lines 17–20 update all debaters' policies. The coalition structure determines which agents share information for cooperative gradient corrections per [**?**].*

# 9 Discussion

## 9.1 Relation to Prior Work

Irving et al. [**?**] established the game-theoretic foundation for debate as an alignment technique, showing that in the idealized setting, the Nash equilibrium of debate is truth-telling. Our contribution is to identify the conditions under which this idealized result degrades: bounded judge rationality (**??**), heterogeneous debater quality (**??**), and—most critically—the self-knowledge bottleneck (**??**).

Khan et al. [**?**] demonstrated empirically that debate can improve the accuracy of human judges on question-answering tasks. Our framework explains their finding: their experimental debaters (language models prompted to argue) have moderate $I_{\text{self}}$ on factual questions (they can often retrieve and articulate the relevant evidence) but lower $I_{\text{self}}$ on reasoning-intensive tasks (where the model's "reasoning" may be pattern-matching).

Bowman et al. [**?**] articulated the scalable oversight problem: how to supervise AI systems more capable than their overseers. Our **??** provides a necessary condition for debate-based oversight: the model must have self-knowledge sufficient to produce honest arguments. This connects to the broader alignment research program by suggesting that *interpretability and debate-based oversight are complementary*: interpretability tools increase $I_{\text{self}}$, making debate more reliable.

## 9.2 Limitations

Several idealizations limit the current analysis:

1. **Binary claims.** Real debates concern complex, multi-dimensional claims. Extension to continuous or structured claim spaces is straightforward but notationally heavier.

2. **Static evidence weights.** We treat $V_i$ as exogenous. In practice, debaters may strategically manipulate their apparent evidence quality (e.g., by making safe, verifiable claims on unimportant points to build track record, then exploiting accrued credibility on the key claim). Modeling this strategic evidence manipulation is an important extension.

3. **Honest judge.** We assume the judge sincerely maximizes accuracy. A corrupt or biased judge introduces a fourth player whose incentives misalign with the truth, requiring extension to adversarial judge models.

4. **Known sparsity.** The blessing-of-dimensionality result assumes the judge knows (or can estimate) the sparsity $k$. Adaptive sparsity estimation is a natural next step.

## 9.3 Conclusion

The $\Omega$-framework—developed for multi-agent policy gradient convergence—provides a surprisingly precise language for the AI safety via debate paradigm. The five components (evidence weighting, opponent shaping, cooperative communication, self-knowledge, and the blessing of dimensionality) each illuminate a different aspect of why debate works, when it fails, and how to improve it.

The central message for AI safety research is a caution: *debate is not a universal oversight mechanism.* It works for models with self-knowledge and fails for models that are vibing. As AI systems grow more capable, ensuring they maintain (or develop) self-knowledge is not just an interpretability desideratum—it is a prerequisite for the most promising alignment techniques.

## References


[1] G. Irving, C. Christiano, and D. Amodei. AI safety via debate. *arXiv preprint arXiv:1805.00899*, 2018.

[2] A. Khan, J. Hughes, D. Valentine, L. Ruis, K. Sachan, A. Raber, R. Guo, H. Iber, G. Irving, and S. R. Bowman. Debating with more persuasive LLMs leads to more truthful answers. *arXiv preprint arXiv:2402.06782*, 2024.

[3] S. R. Bowman, J. Hyun, E. Perez, E. Chen, C. Pettit, S. Heiner, K. Lukosuite, A. Askell, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinnon, C. Olah, D. Amodei, D. Amodei, D. Drain, D. Li, E. Tran-Gyenes, et al. Measuring progress on scalable oversight for large language models. *arXiv preprint arXiv:2211.03540*, 2022.

[4] A. Giannou, K. Lotidis, P. Mertikopoulos, and E.-V. Vlatakis-Gkaragkounis. On the convergence of policy gradient methods to Nash equilibria in general stochastic games. *arXiv preprint arXiv:2210.08857*, 2022.

[5] E. Shcherbinin. Evidence-weighted policy gradient convergence in general stochastic games. Working paper, LSE, 2026.

[6] E. Shcherbinin. Cooperative policy gradient with lossy communication in general stochastic games. Working paper, LSE, 2026.

[7] E. Shcherbinin. The blessing of dimensionality for policy gradients: sparse recovery, evidence weighting, and the logarithmic cost of action space expansion. Working paper, LSE, 2026.

[8] R. Parvizi-Wayne, S. Kotler, G. Mannino, and K. Friston. Forgetting ourselves in flow: an active inference account of flow states. *Frontiers in Psychology*, 15:1354719, 2024.

[9] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.

[10] E. J. Candès, J. K. Romberg, and T. Tao. Stable signal recovery from incomplete and inaccurate measurements. *Communications on Pure and Applied Mathematics*, 59(8):1207–1223, 2006.